

COMS3021A Test 1

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This test contributes 5% to your final mark

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1. (5 Points) Provide the equivalent language for the computational problem of:

Given a binary natural number $\langle n \rangle_{\mathbb{N},2}$, compute the size of the language of all binary strings of length n .

- Computational Problem

$$\text{FIXED_STRING_COUNT} : \{0, 1\}^* \mapsto \{0, 1\}^*$$

$$\text{FIXED_STRING_COUNT}(\langle n \rangle_{\mathbb{N},2}) = \langle |\{0, 1\}^n| \rangle_{\mathbb{N},2}$$

- Equivalent Decision Problem

$$\text{IS_FIXED_STRING_COUNT} : \{0, 1\}^* \mapsto \{0, 1\}$$

$$\text{IS_FIXED_STRING_COUNT}(\langle n, m \rangle_{\mathbb{N},2}) = \begin{cases} 1 & \text{if } m = |\{0, 1\}^n| \\ 0 & \text{if } m \neq |\{0, 1\}^n| \end{cases}$$

- ▶ Given the binary represented natural number n and m , decide if m is the number of strings in the language $\{0, 1\}^n$.

- Equivalent Language

$$L = \left\{ v \in \{0, 1\}^* \mid v = \langle n, |\{0, 1\}^n| \rangle_{\mathbb{N},2} \right\}$$

- ▶ The set of binary represented natural numbers n and $|\{0, 1\}^n|$
- ▶ +2 Marks: Correct problem / elements in correct form
- ▶ +1 Mark: is a language $L \subseteq \Sigma^*$
- ▶ +1 Mark: is a binary language $L \subseteq \{0, 1\}^*$
- ▶ +1 Mark: Language size is a natural number

2. (5 Marks) Provide a binary representation for the following data structure:

A rational polynomial $p(x)$ is defined as the sum of rational coefficients and decreasing natural number powers of x

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

- +1 Mark: Provided $\langle a_i x^i \rangle$
- +2 Marks: $\langle p(x) \rangle$ provided and powers are appropriately handled (why has been left out etc.)
- +2 Marks: Specifies a binary encoding / natively binary representation

For all below answers:

$$e(x) = \begin{cases} 00 & \text{if } x = 0 \\ 01 & \text{if } x = 1 \\ 10 & \text{if } x = \# \\ 11 & \text{if } x = ; \end{cases}$$

$$\langle a_k x^k \rangle = \langle a_k \rangle_{\mathbb{Q},2}$$

$$\langle p(x) \rangle = \langle a_n x^n \rangle \# \langle a_{n-1} x^{n-1} \rangle \# \dots \# \langle a_2 x^2 \rangle \# \langle a_1 x^1 \rangle \# \langle a_0 x^0 \rangle$$

The powers of x do not need to be encoded as we map the n -th power to the n -th coefficient.

- The k th coefficient in the list is has k as its power
 - We are enforcing an ordering on the coefficients.
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$$\langle a_k x^k \rangle = \langle a_k \rangle_{\mathbb{Q},2} \# \langle k \rangle_{\mathbb{N},2}$$

$$\langle p(x) \rangle = \langle a_0 x^0 \rangle; \langle a_1 x^1 \rangle; \langle a_2 x^2 \rangle; \dots; \langle a_n x^n \rangle$$

We do not impose an ordering on $\langle p(x) \rangle$, if we receive $\langle a_1 x^1 \rangle \# \langle a_2 x^2 \rangle \# \langle a_0 x^0 \rangle$ we can still recover the original polynomial.